

# Mathematical Framework and Calibration of the Heston Stochastic Volatility Model

## Quantitative Research Working Paper

May 24, 2026

### Abstract

This paper outlines the mathematical foundations of the Heston Stochastic Volatility Model, addressing the empirical limitations of the constant-volatility Black-Scholes framework. We detail the system of coupled Stochastic Differential Equations (SDEs), discuss the asset price dynamics under the risk-neutral measure, provide the analytical solution via Fourier transform inversion of the characteristic function, and formulate the non-linear optimization routine required to calibrate the model parameters  $(\kappa, \theta, \sigma, \rho, v_0)$  to live options market data.

## 1 Introduction and Empirical Motivation

The foundational Black-Scholes-Merton model assumes that the volatility of the underlying asset is a constant parameter  $\sigma$ . Empirically, this assumption is flatly contradicted by options market data. When plotting the implied volatility calculated from market prices against strike prices for a fixed maturity, a distinct U-shaped or smirk-like pattern emerges, known as the *Volatility Smile*. This phenomenon reflects the market's pricing of tail risk and skewness, driven by investor risk aversion and leverage effects.

To resolve this discrepancy, stochastic volatility models treat variance itself as a latent, diffusion-driven process. The Heston model (1993) remains one of the most prominent frameworks because it provides a semi-analytical closed-form solution for European options, enabling computationally efficient calibration.

## 2 The Core Mathematical Framework

The Heston model simulates the joint dynamic behavior of an asset price  $S_t$  and its variance  $v_t$  under the risk-neutral pricing measure  $\mathbb{Q}$  using the following system of coupled Stochastic Differential Equations (SDEs):

$$dS_t = rS_t dt + \sqrt{v_t}S_t dW_t^1 \quad (1)$$

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t^2 \quad (2)$$

where  $W_t^1$  and  $W_t^2$  are two standard Brownian motions defined on a filtered probability space, possessing a constant instantaneous correlation:

$$d\langle W^1, W^2 \rangle_t = \rho dt \quad (3)$$

The structural parameters governing this bivariate diffusion system are defined as follows:

- $r \in \mathbb{R}$ : The risk-free continuous interest rate.

- $v_t \geq 0$ : The instantaneous variance at time  $t$ , initializing at  $v_0$ .
- $\theta > 0$ : The long-term mean variance (the asymptotic value toward which the variance reverts).
- $\kappa > 0$ : The rate or speed of mean reversion.
- $\xi > 0$ : The volatility of volatility, determining the variance of the variance process.
- $\rho \in [-1, 1]$ : The correlation coefficient between the asset price shocks and variance shocks. A negative value ( $\rho < 0$ ) captures the empirical *leverage effect*, where an asset price drop triggers an immediate spike in market volatility.

To guarantee that the variance process  $v_t$  remains strictly positive and never touches zero, the model parameters must satisfy the **Feller condition**:

$$2\kappa\theta > \xi^2 \quad (4)$$

### 3 Analytical Pricing via Fourier Inversion

Under risk-neutral valuation theory, the price of a European call option  $C(S_t, v_t, t; K, T)$  with strike price  $K$  and maturity  $T$  is given by the discounted expected payoff:

$$C = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} [\max(S_T - K, 0) \mid \mathcal{F}_t] \quad (5)$$

By analogy to the Black-Scholes formula, Heston framed the solution using two conditional probabilities:

$$C = S_t P_1 - K e^{-r(T-t)} P_2 \quad (6)$$

where  $P_1$  represents the right-to-exercise probability weighted by the asset price, and  $P_2$  is the standard conditional probability of the option expiring in-the-money under the risk-neutral measure.

Because the conditional distribution cannot be easily derived in terms of standard probability density functions, we solve for  $P_1$  and  $P_2$  by taking the inverse Fourier transform of their respective characteristic functions  $f_j(x, v_t, \tau; u)$ :

$$P_j(x, v_t, \tau; \ln K) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[ \frac{e^{-iu \ln K} f_j(x, v_t, \tau; u)}{iu} \right] du \quad (7)$$

where  $\tau = T - t$  is the time-to-maturity,  $x = \ln S_t$ , and the characteristic functions are expressed exponentially as:

$$f_j(x, v_t, \tau; u) = \exp \{ C_j(\tau, u) + D_j(\tau, u) v_t + i u x \} \quad (8)$$

The coefficients  $C_j$  and  $D_j$  are solved from a system of partial differential equations derived from the Feynman-Kac theorem:

$$C_j(\tau, u) = r i u \tau + \frac{\kappa \theta}{\xi^2} \left[ (b_j - \rho \xi i u + d_j) \tau - 2 \ln \left( \frac{1 - g_j e^{d_j \tau}}{1 - g_j} \right) \right] \quad (9)$$

$$D_j(\tau, u) = \frac{b_j - \rho \xi i u + d_j}{\xi^2} \left( \frac{1 - e^{d_j \tau}}{1 - g_j e^{d_j \tau}} \right) \quad (10)$$

where the internal algebraic variables are given by:

$$g_j = \frac{b_j - \rho \xi i u + d_j}{b_j - \rho \xi i u - d_j} \quad (11)$$

$$d_j = \sqrt{(b_j - \rho \xi i u)^2 - \xi^2 (2u_j i u - u^2)} \quad (12)$$

For  $j = 1$ :  $b_1 = \kappa - \rho \xi$ ,  $u_1 = \frac{1}{2}$ .  
For  $j = 2$ :  $b_2 = \kappa$ ,  $u_2 = -\frac{1}{2}$ .

## 4 Model Calibration as an Optimization Problem

To utilize the Heston pricing model in a production environment, the latent parameter vector  $\Phi = \{v_0, \kappa, \theta, \xi, \rho\}$  must be calibrated against observed market data. Let  $N$  be the number of liquid European options traded in the market for a given asset, with strikes  $K_i$  and maturities  $T_i$ .

Let  $C^{\text{mkt}}(K_i, T_i)$  represent the mid-market price observed in reality, and let  $C^{\text{Heston}}(S_0, K_i, T_i; \Phi)$  represent the price generated by our Fourier integration engine. The objective is to find the optimal parameter set  $\Phi^*$  that minimizes a loss function, typically chosen as the Mean Squared Error (MSE):

$$\Phi^* = \arg \min_{\Phi} \frac{1}{N} \sum_{i=1}^N \left( C^{\text{Heston}}(S_0, K_i, T_i; \Phi) - C^{\text{mkt}}(K_i, T_i) \right)^2 \quad (13)$$

This problem is non-linear and non-convex, exhibiting multiple local minima. Computational optimization algorithms such as Nelder-Mead (a simplex-based direct search method) or the Levenberg-Marquardt algorithm are utilized to iterate through candidate states, penalizing parameter vectors that violate boundary constraints or fail the Feller condition ( $2\kappa\theta < \xi^2$ ).

## 5 Conclusion

The Heston model elegantly captures structural market features including time-varying volatility clustering, mean reversion, and asymmetry due to price-volatility correlations. By mapping out a 3D Volatility Surface through inverse Fourier transformations and optimization loops, quantitative analysts establish a mathematically consistent framework to price exotic derivatives, manage portfolio risk, and discover alpha across global option chains.